

game will notice that we're not working with the exact shape of each game element. We'll treat the ship and the screen as rectangles in this class.

We import the `pygame` module before defining the class. The `__init__()` method of `Ship` takes two parameters: the `self` reference and a reference to the current instance of the `AlienInvasion` class. This will give `Ship` access to all the game resources defined in `AlienInvasion`. We then assign the screen to an attribute of `Ship` ❶, so we can access it easily in all the methods in this class. We access the screen's `rect` attribute using the `get_rect()` method and assign it to `self.screen_rect` ❷. Doing so allows us to place the ship in the correct location on the screen.

To load the image, we call `pygame.image.load()` ❸ and give it the location of our ship image. This function returns a surface representing the ship, which we assign to `self.image`. When the image is loaded, we call `get_rect()` to access the ship surface's `rect` attribute so we can later use it to place the ship.

When you're working with a `rect` object, you can use the *x*- and *y*-coordinates of the top, bottom, left, and right edges of the rectangle, as well as the center, to place the object. You can set any of these values to establish the current position of the `rect`. When you're centering a game element, work with the center, `centerx`, or `centery` attributes of a `rect`. When you're working at an edge of the screen, work with the top, bottom, left, or right attributes. There are also attributes that combine these properties, such as `midbottom`, `midtop`, `midleft`, and `midright`. When you're adjusting the horizontal or vertical placement of the `rect`, you can just use the *x* and *y* attributes, which are the *x*- and *y*-coordinates of its top-left corner. These attributes spare you from having to do calculations that game developers formerly had to do manually, and you'll use them often.

NOTE

In Pygame, the origin (0, 0) is at the top-left corner of the screen, and coordinates increase as you go down and to the right. On a 1200×800 screen, the origin is at the top-left corner, and the bottom-right corner has the coordinates (1200, 800). These coordinates refer to the game window, not the physical screen.

We'll position the ship at the bottom center of the screen. To do so, make the value of `self.rect.midbottom` match the `midbottom` attribute of the screen's `rect` ❹. Pygame uses these `rect` attributes to position the ship image so it's centered horizontally and aligned with the bottom of the screen.

Finally, we define the `blitme()` method ❺, which draws the image to the screen at the position specified by `self.rect`.

Drawing the Ship to the Screen

Now let's update `alien_invasion.py` so it creates a ship and calls the ship's `blitme()` method:

```
alien
_invasion.py
--snip--
from settings import Settings
from ship import Ship
```